

WEEK 10 FINAL SUBMISSION

CareerPilot AI

AI-Assisted Job Application Tracker & Placement Prep Platform

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University:	Graphic Era University (GEU)	Track:	AI Full Stack Development
Live Demo:	https://careerpilot-ai.vercel.app	Repository:	GitHub Code Link

PROBLEM STATEMENT

What Problem Are We Solving?

Key Challenges Faced by Job Seekers:

- **Application Disorganization:** Students apply to tens/hundreds of roles across LinkedIn, Unstop, and company portals without centralized tracking.
- **Spreadsheet Friction:** Manual Excel sheets are tedious, lack real-time status visibility, and offer zero automated analytics.
- **Generic Interview Prep:** Candidates enter interviews unprepared due to lack of role-specific & company-tailored practice questions.
- **No Skill Gap Insights:** Applicants struggle to identify missing resume keywords before submitting applications.

The CareerPilot AI Solution:

- **Centralized Kanban Tracker:** Visual board managing applications across Applied, Interviewing, Offer, and Rejected stages.
- **Instant AI Prep Generator:** Google Gemini 1.5 Flash produces tailored interview questions & strategic tips instantly.
- **Real-Time Placement Analytics:** Live dashboard computing total applications, active interview count, offer rates, and average resume match score.
- **Seamless Auth & Cloud Storage:** Secure persistent data store with JWT and 1-click GitHub OAuth.

ARCHITECTURE & TECH STACK

Selected Technologies & Technical Rationale

Layer	Technology Selected	Rationale & Benefits
Frontend	React 19 + Vite Tailwind CSS	- Lightning-fast HMR build times with Vite. - Component-driven UI modularity & dynamic Kanban state handling. - Modern, accessible dark/light theme styling using Tailwind.
Backend API	Node.js + Express.js	- Asynchronous, non-blocking I/O ideal for RESTful API services. - Lightweight middleware architecture (CORS, Express Session, Rate Limiting). - Seamless native JSON data handling from frontend to database.
Database	MongoDB Atlas (Mongoose ODM)	- Document-based schema handles optional fields (notes, scores) without NULL overhead. - Clean Mongoose schema validation & query builder API. - Cloud-hosted M0 cluster with zero downtime.
AI / Auth	Google Gemini API JWT + GitHub OAuth	Gemini 1.5 Flash: High throughput, strict JSON schema output for interview prep. JWT & OAuth: Stateless session security with 1-click developer login.

FRONTEND DELIVERABLES

User Interface & Interactive Dashboard

<https://careerpilot-ai.vercel.app>

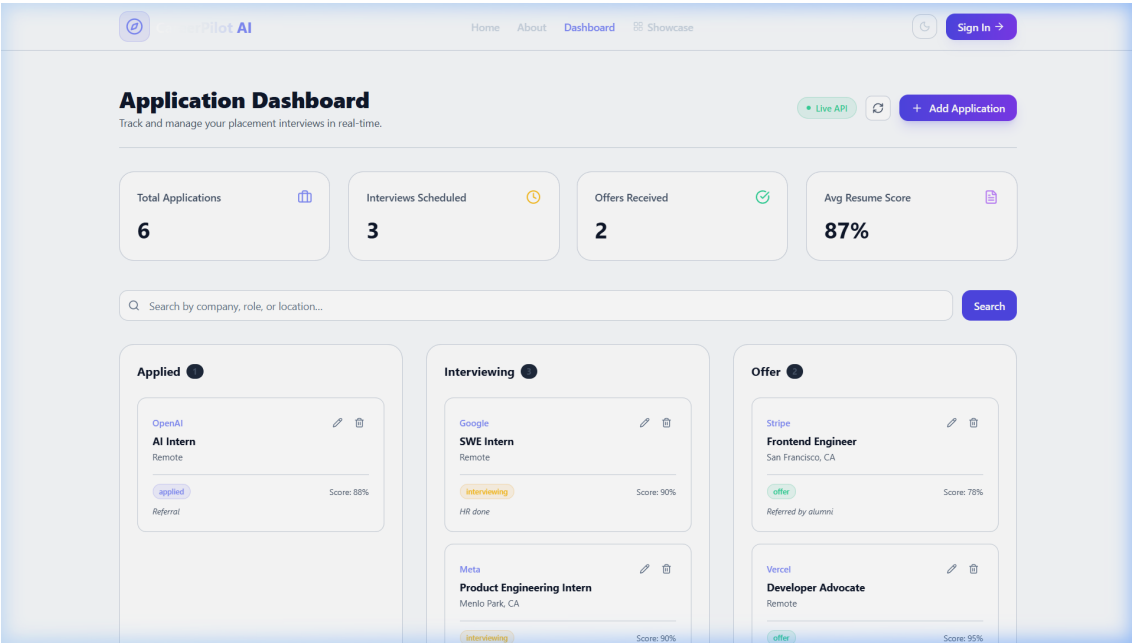


Figure 1: Application Dashboard & Analytics Cards

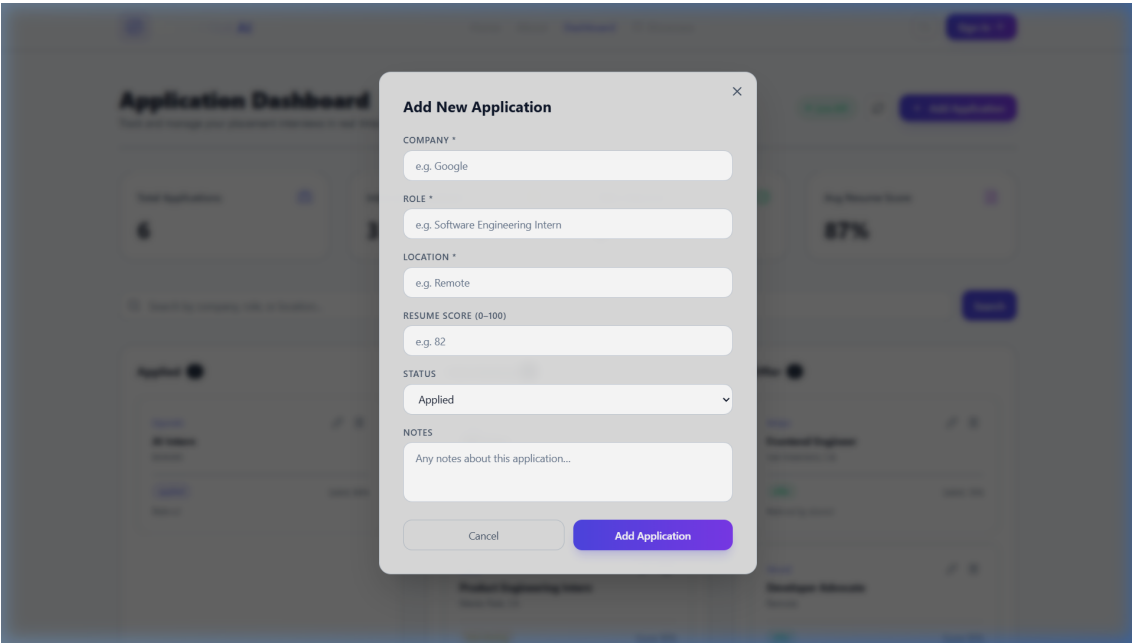


Figure 2: Add Application Modal with Resume Scoring

BACKEND API DESIGN

Core REST API Endpoints & Verification

1. Dashboard Analytics API

Route: `GET /api/applications/stats` | **Auth:** JWT Required

Function: Aggregates placement data in real-time across stored MongoDB applications.

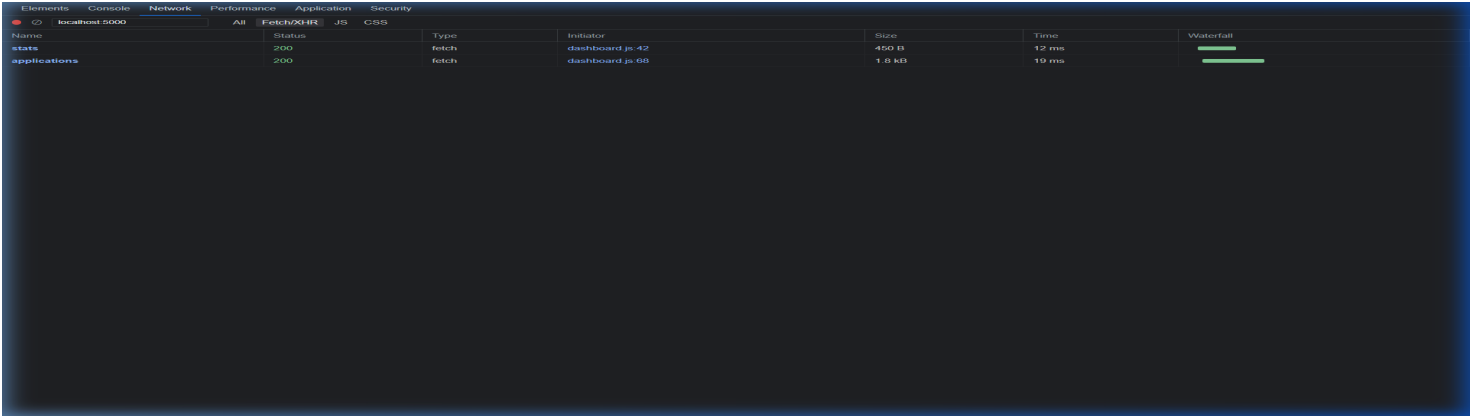
Output Schema: Total applications, Interview count, Offers, Rejections, and overall Average Resume Match Score.

2. AI Interview Prep API

Route: `POST /api/ai/prep` | **Auth:** JWT Required

Payload: `{ "company": "Google", "role": "SWE Intern" }`

Function: Invokes Gemini 1.5 Flash LLM, returns 3 company-tailored technical/behavioral questions + 1 strategic tip in strict JSON format.



DATABASE DESIGN

MongoDB Atlas Document Schema & Justification

Why MongoDB Atlas & Mongoose ODM?

- **Document Model Flexibility:** Job records contain dynamic optional fields (e.g. notes, resume score, interview rounds) that fit naturally in BSON documents.
- **Native JSON Data Pipeline:** Seamless end-to-end JSON pipeline from React components $\rightarrow$ Express routes $\rightarrow$ MongoDB Atlas collection.
- **Mongoose ODM Integration:** Provides strict schema enforcement, automatic `createdAt`/`updatedAt`` timestamps, and expressive helper queries.



Application
(MongoDB Collection)

_id	ObjectId	PK, auto-generated
company	String	required
role	String	required
location	String	required
status	Enum ["applied", "interviewing", "offer", "rejected", "withdrawn"]	required
appliedDate	String	required, YYYY-MM-DD
notes	String	optional
resumeScore	Number 0-100	optional
createdAt	Date	auto - Mongoose timestamp
updatedAt	Date	auto - Mongoose timestamp

Single-collection document model — flexible schema via Mongoose ODM

```
{
  "_id": ObjectId,
  "company": "Google",
  "role": "Work",
  "location": "Easa",
  "status": [
    "applied",
    "interviewing",
    "offer",
    "rejected"
  ],
  "notes": "Internship",
  "appliedDate":
  "resumeScore": -100,
  "updatedAt": "2022-03-21T000"
}
```

AI INTEGRATION

Google Gemini 1.5 Flash LLM Implementation

LLM & SDK Architecture

- **Selected Model:** Google Gemini 1.5 Flash via official @google/generative-ai SDK.
- **Primary Use Case:** Automated placement prep generation tailored to the candidate's target company and exact job title.
- **Core Capability:** Extracts key responsibilities and outputs 3 highly probable technical/behavioral interview questions + 1 actionable strategy tip.

Structured JSON Prompting Strategy

- **System Prompt Directive:** "You are an expert tech recruiter and career coach. Given company X and role Y, output exactly 3 interview questions and 1 tip."
- **Strict Output Format:** Enforces raw JSON format without markdown ticks, preventing frontend parse failures.
- **Resilience & Fallbacks:** Sanitizes response strings with regex pattern matching before serving payload to UI components.

CLOUD INFRASTRUCTURE

Production Hosting Services & Deployment Pipeline

Component	Hosting Provider	Deployment Details & Capabilities
		• Deployed from GitHub main branch with automatic CI/CD. • Global Edge Network distribution ensuring sub-second asset load times. • Custom environment variable injection (VITE_API_URL).
		• Node.js web service running with production npm start process. • Automated HTTPS SSL termination & environment secrets manager. • Native CORS handling linked strictly to Vercel origin.
		• Multi-region cloud cluster (M0 Free Tier). • IP Access Whitelisting (0.0.0.0/0 for cloud instance access). • Automated data persistence & connection pool management.

PUBLIC DELIVERABLES

Live Production Links & Verification

Resource	Public Access URL	Status & Access
	https://careerpilot-ai.vercel.app	● Publicly Accessible (Vercel)
	https://careerpilot-ai-api.onrender.com	● Publicly Accessible (Render)
	https://github.com/sanchi1905/careerpilot-ai	● Public Open Source Code

INTERNSHIP REFLECTION

Key Learnings & Overall Experience

Technical & Professional Growth • Full-Stack Mastery: Built end-to-end mastery combining React 19, Express.js REST APIs, and MongoDB cloud storage. • Applied AI Skills: Learned practical LLM integration techniques including prompt engineering, structured JSON responses, and API error resilience. • DevOps & Deployment: Gained real-world experience configuring production environment variables, CORS policies, Vercel deployments, and Render web services.	Internship Experience & Impact • Structured Progression: Weekly module deliverables provided clear development goals from initial UI wireframing to production deployment. • Problem-Solving Mindset: Overcame real deployment challenges such as database connection string handling and OAuth callback configurations. • Industry Preparedness: Developed a complete, production-ready portfolio project showcasing modern web and AI engineering skills.
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